

RESEARCH EXPERIENCES

University of Illinois Chicago

Chicago, IL, USA

Research Area: RL, Reasoning, Causal RL, Multi-Agent Learning, Network Structure Inference, LLMs

Jan 2026 – Present

- **Project 1: Built a VERL-based RL framework to break i.i.d. sampling** to encourage a more uniform output distribution in LLM generation through **set-level optimization**.
- Developed a **GRPO training pipeline** with a custom **set-level reward function** to reduce genre collapse and improve distributional balance across generated outputs.
- Integrated **vLLM for rollout**, **SLIME** to **decouple rollout** from training input pipelines, **SGLang** to improve **high-throughput inference** orchestration, and **Megatron** for distributed model weight updates on a **multi-GPU SLURM/Delta cluster**, optimizing GPU utilization and improving large-scale training efficiency.
- This work is under review at **AAAI, 2026**.
- **Project 2: Causal Reinforcement Learning for Robotics Control**
- Developed a **causal reinforcement learning framework** for continuous-control **robotics tasks** by integrating **causal discovery**, **counterfactual data augmentation**, and **off-policy policy optimization** into a SAC-based learning pipeline for simulated robot control.
- Applied causal discovery methods to learn state-action-reward dependency structures and **evaluate causal identifiability** under nonlinear environment dynamics.
- Designed **simulator-consistent counterfactual trajectory** generation mechanisms for causal intervention analysis and synthetic experience augmentation.
- Conducted **large-scale ablation studies** on causal discovery, counterfactual augmentation, and policy optimization components, improving episodic return by over 40% on MuJoCo HalfCheetah benchmarks.
- **Project 3: Construct user-level interaction/exposure signals from social context (e.g., neighbor/peer information flow), using LLM-based text representations to study how network influence relates to stance switching.**
- Define **RL-style objectives for stance dynamics** (e.g., stability vs. switching, cost/constraints) to support **counterfactual evaluation of sequential interventions**.
- Model users as agents in a networked **multi-agent setting**, where each agent's behavior can be influenced by neighbors, aligning the task with **causal RL under spillovers**.

Northwestern University (On-site)

Chicago, IL, USA

Researcher (Network Analysis, NLP, LLMs)

May 2025 – Dec 2025

- Developed **NLP and representation-learning pipelines** using SciBERT, topic modeling, and citation networks to characterize scientific field evolution, identify knowledge gaps, and discover emerging research opportunities.
- Applied **network analytics** and causal discovery techniques to study mechanisms underlying **scientific novelty** and domain evolution; co-author of “Measuring Scientific Novelty as Network Rewiring” (under review, NetSci 2026).

Greaten AI (Engineering and Research Team, Part-time, Remote)

Albany, CA, USA

Software Development Engineer (Multi-Agent, Reinforcement Learning, LLMs)

Dec 2024 – May 2025

- Developed an adaptive LLM multi-agent framework with **reinforcement learning–based topology optimization**, using a **LoRA-tuned graph selector** to dynamically determine agent communication structures across reasoning, QA, and code-generation tasks.
- Built and deployed **production-grade RAG** and **voice-agent systems**, integrating multimodal document understanding, retrieval-augmented generation, AWS telephony services, and containerized infrastructure for scalable, low-latency enterprise applications.

University of Chicago (Part&Full-time, On-site)

Chicago, IL, USA

Research Assistant (Bioscience, Causal Analysis, Biomarker Analysis)

Jan 2024 – Feb 2025

- Conducted **biostatistical analyses** on large-scale environmental, microbiome, and biomarker datasets to quantify the direct and indirect effects of PM2.5 exposure on cardiometabolic health outcomes.
- Developed robust mediation-analysis pipelines incorporating propensity score matching, bootstrap resampling, and multiple-testing correction to identify biologically meaningful **exposure pathways**.

WORK EXPERIENCES

OOIN Media (Full-time, On-site)

Long Island, NY, USA

SDE-AI Agent Engineer

Aug 2025 – Jan 2026

- Architected a production-scale **multi-agent system with LangGraph**, hierarchical planning, **multi-LLM routing**, and **tool orchestration** to automate **merchant–creator collaboration workflows**, including creator discovery, matching, outreach, negotiation, and campaign management.
- Developed an AI-powered video generation system combining trend analysis, content planning, and **iterative refinement** to improve content quality and consistency.

Hillhouse Capital (Big Health, Full-time, On-site)

Beijing, China

Senior Machine Learning Engineer

May 2022 – Jul 2023

- Built an end-to-end healthcare **recommendation platform** combining two-tower retrieval networks, representation learning, and LLM-based content embeddings, supported by Spark-powered real-time inference pipelines and validated through production A/B testing with a 2% conversion uplift.

DiDi (User Growth Department, Full-time, On-site)

Beijing, China

Senior Machine Learning Engineer

Sep 2020 – May 2022

- Developed a deep machine learning framework for marketing intervention optimization, combining **multi-arm bandit**, heterogeneous treatment effect estimation, and budget-constrained optimization to maximize campaign ROI, resulting in a 20% increase in user adoption and GMV while reducing costs by 85%.

FinUp (Credit Risk Department, Full-time, On-site)

Beijing, China

Model Analyst

Jan 2018 – Sep 2020

- Built end-to-end credit **risk and fraud modeling frameworks** using gradient boosting, statistical learning, and sequence models, and translated model outputs into lending and pricing strategies that reduced overdue rates by 15%.

PUBLICATIONS

- Jiajun Luo, **Yuqing Wu**, *et al.* “The Role of Oral Microbiome in Circulating Metabolic Biomarkers and the Influence of Air Pollution.” *Oral Diseases*, **January 2026**.
- Hui Yi Leong*, **Yuqing Wu***, *et al.* “AMAS: Adaptively Determining Communication Topology for LLM-based Multi-agent System.” **EMNLP of 2025: Industry Track, 2025. (Co-first authors)**

EDUCATION

University of Illinois Chicago, *Doctoral of Computer Science*
University of Chicago, *MS in Applied Data Science, GPA 4.0*

Jan 2026
Sep 2023 – Dec 2024

SKILLS

- **Programming:** Python (TensorFlow, PyTorch), R, SQL, Scala, Java
- **Full-Stack Development:** Frontend (Java, HTML, React, Vite, JavaScript); Bankend (Python, Java, HTTP, TCP)
- **Dev Tools & Platforms:** Spark(PySpark), Hive, HDFS, Docker, RESTful APIs, Flask/FastAPI, Git, Linux, GCP, AWS, CRM system
- **Machine Learning/Deep Learning:** LLM, AI-Agent, Generative AI, RL, RAG, NLP, Recommendation Systems, Causal Inference, Optimization Techniques, Anomaly Detection, Multi-task Learning
- **Analytics:** Product analytics, Marketing strategy, Risk analysis, Supply-demand optimization, Healthcare management
- **Visualization:** Tableau, Power BI, Superset